

DSA1101

Introduction to Data Science

Week 5 Tutorial 3

TODAY'S AGENDA

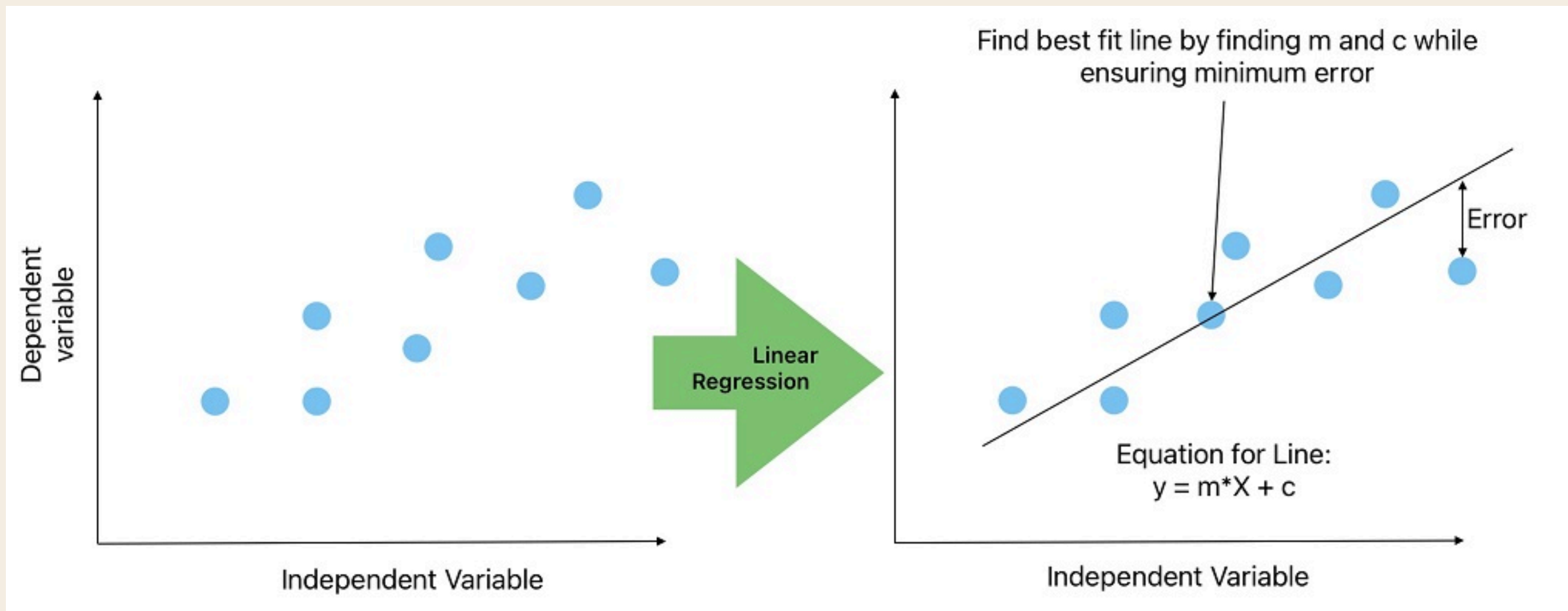
1. Quick Revision
2. On-site Questions Attempt
3. On-site Questions Discussion
4. Off-site Questions Discussion



QUICK REVISION

Linear Regression

A supervised learning algorithm that models the relationship between a continuous response variable and one or more predictors by fitting a linear model (a line, plane, or hyperplane depending on the number of predictors) to the data.



Linear Regression

Assumptions (for this course):

1. Response is quantitative

2. Errors are normally distributed

→ check by examining if **response is approximately symmetric** (transform if needed)

3. Errors have constant variance

→ check if **variability of y is stable** when x changes

Linear Regression

Assumptions (for this course):

1. Response is quantitative

2. Errors are normally distributed

→ check by examining if response is approximately symmetric (transform if needed)

→ plot histogram of response variable

3. Errors have constant variance

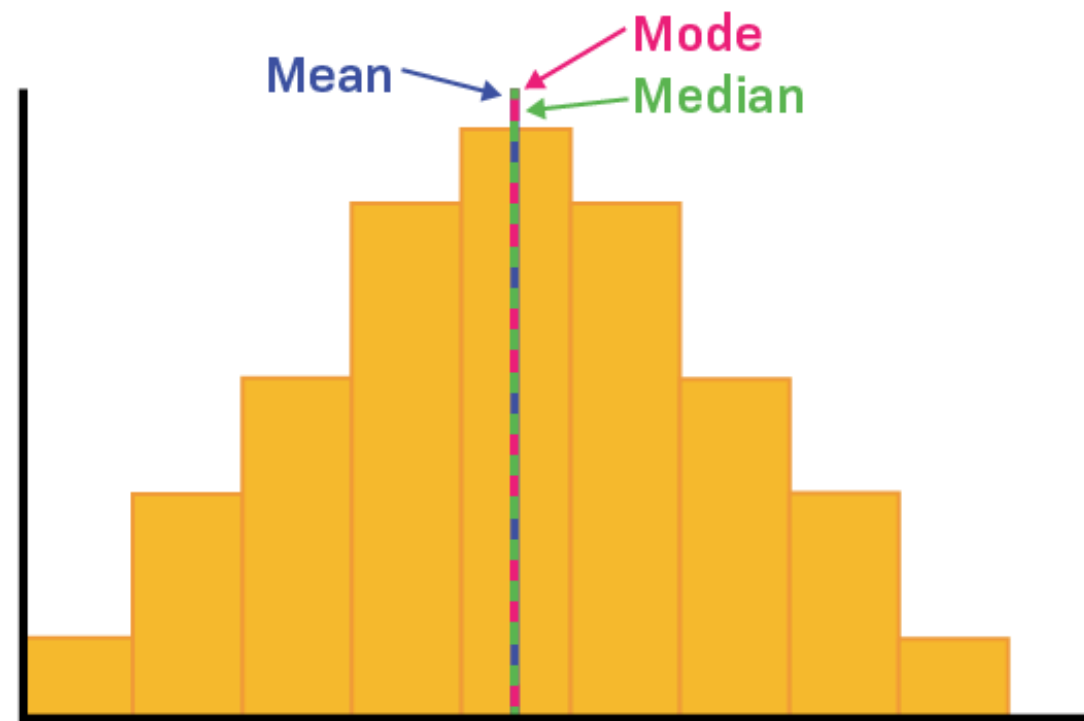
→ check if variability of y is stable when x changes

→ plot scatterplot of y vs x

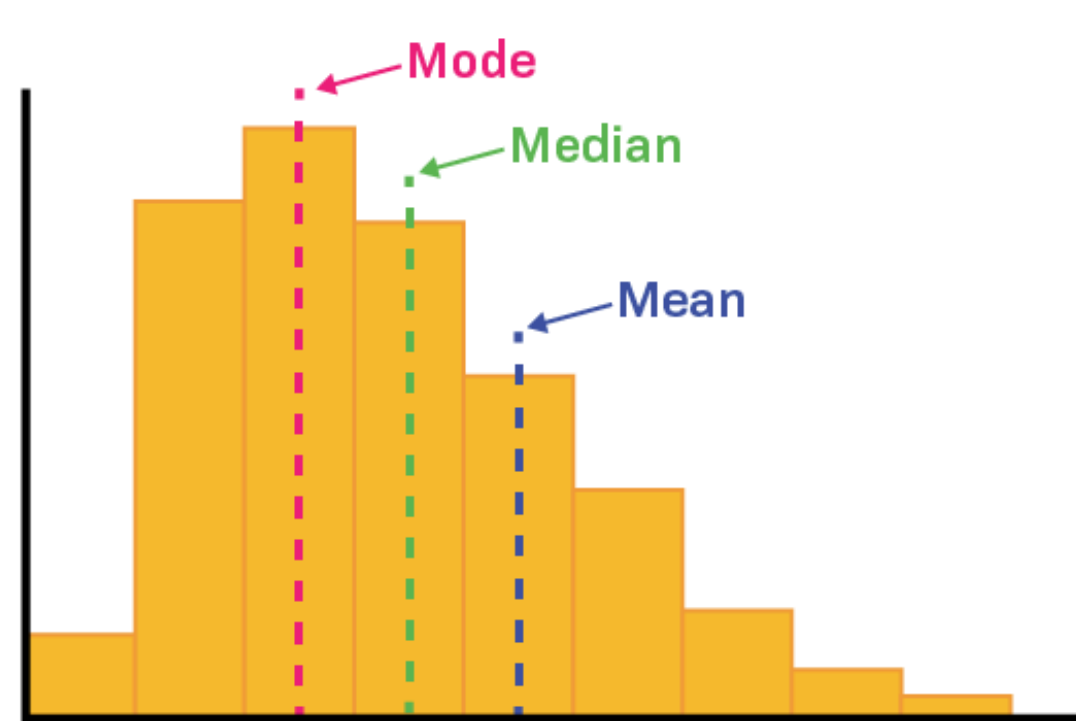
Linear Regression

From last week:

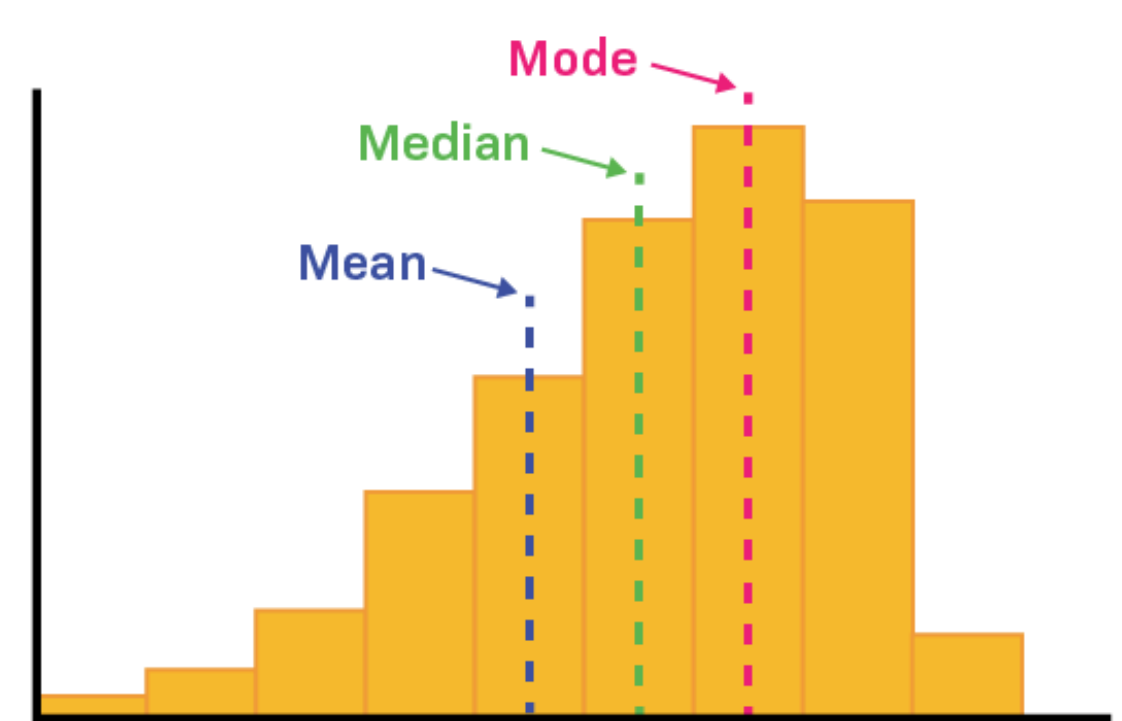
A. Symmetric



B. Right-skewed (or Positive-skewed)

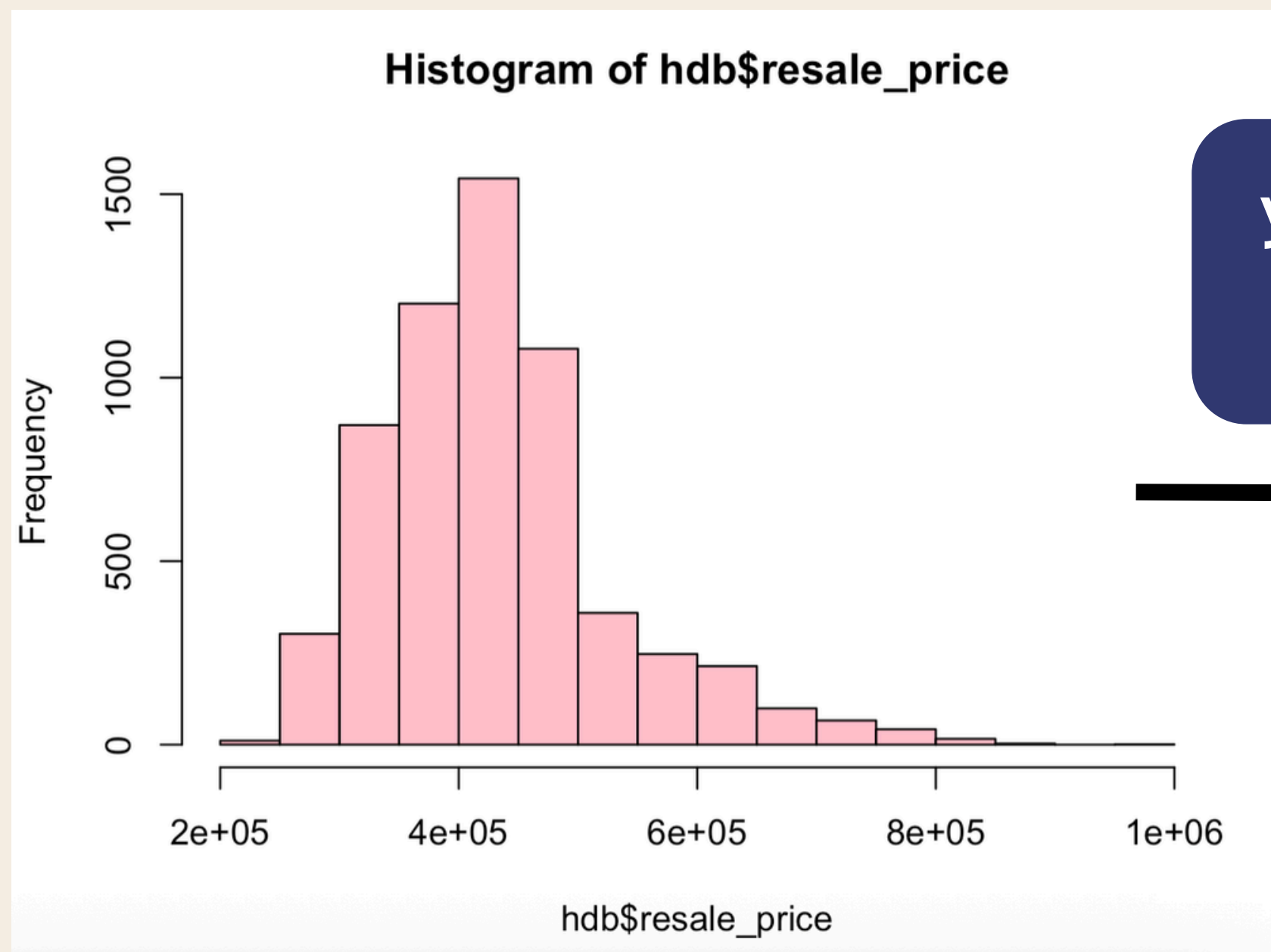


C. Left-skewed (or Negative-skewed)

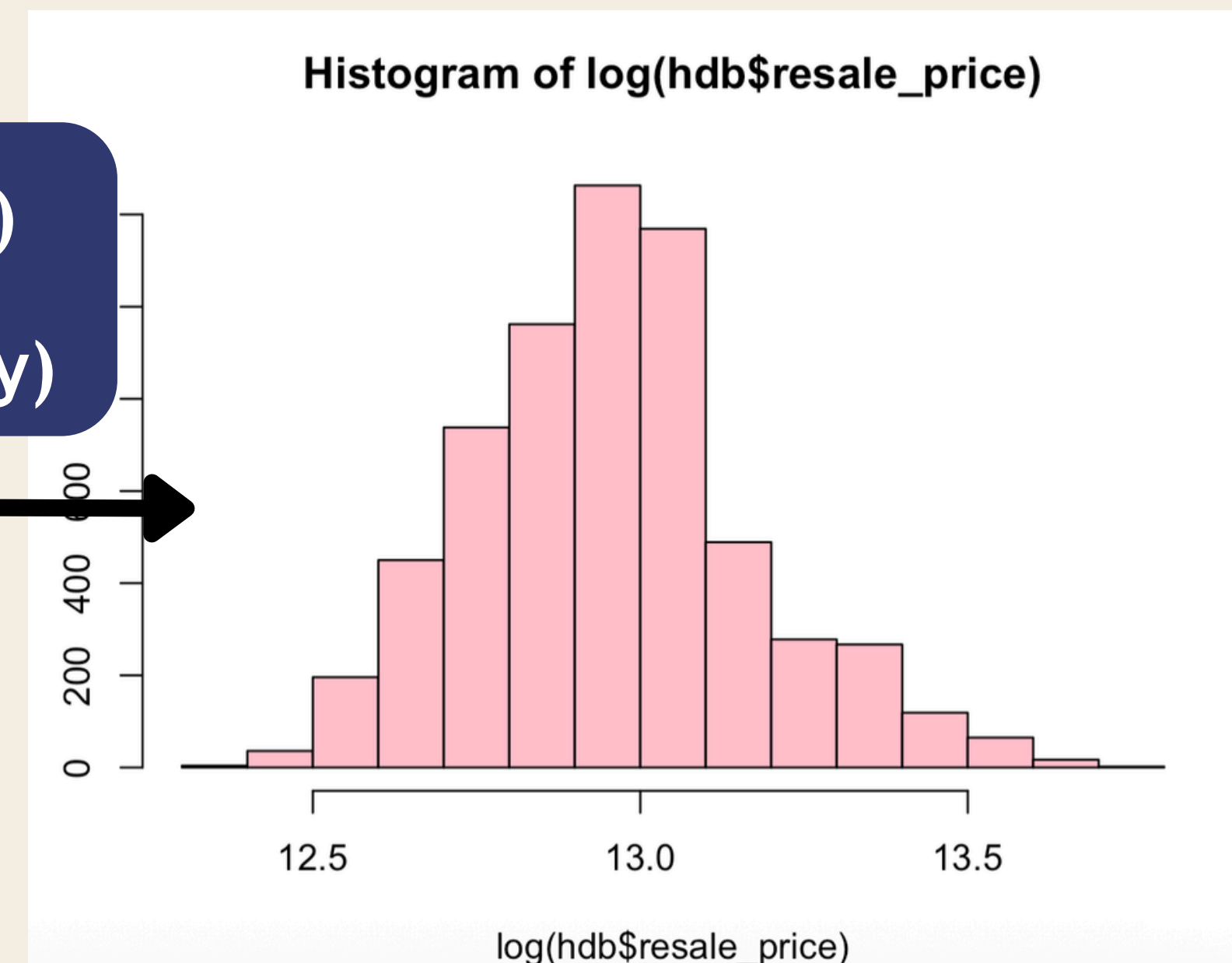


Transforming skewed variables

if right-skewed:

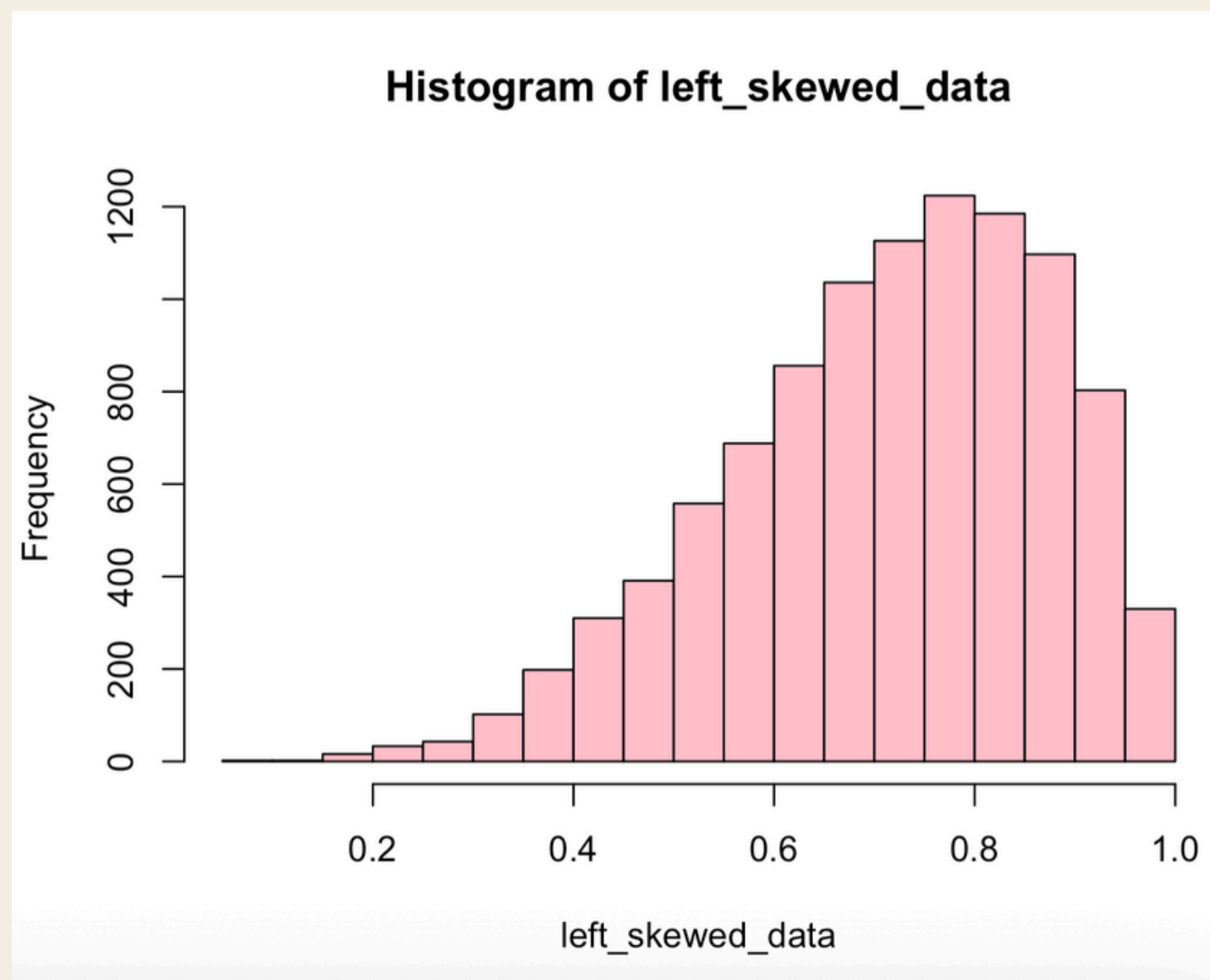


$y \rightarrow \log(y)$
 $\rightarrow \sqrt{y}$

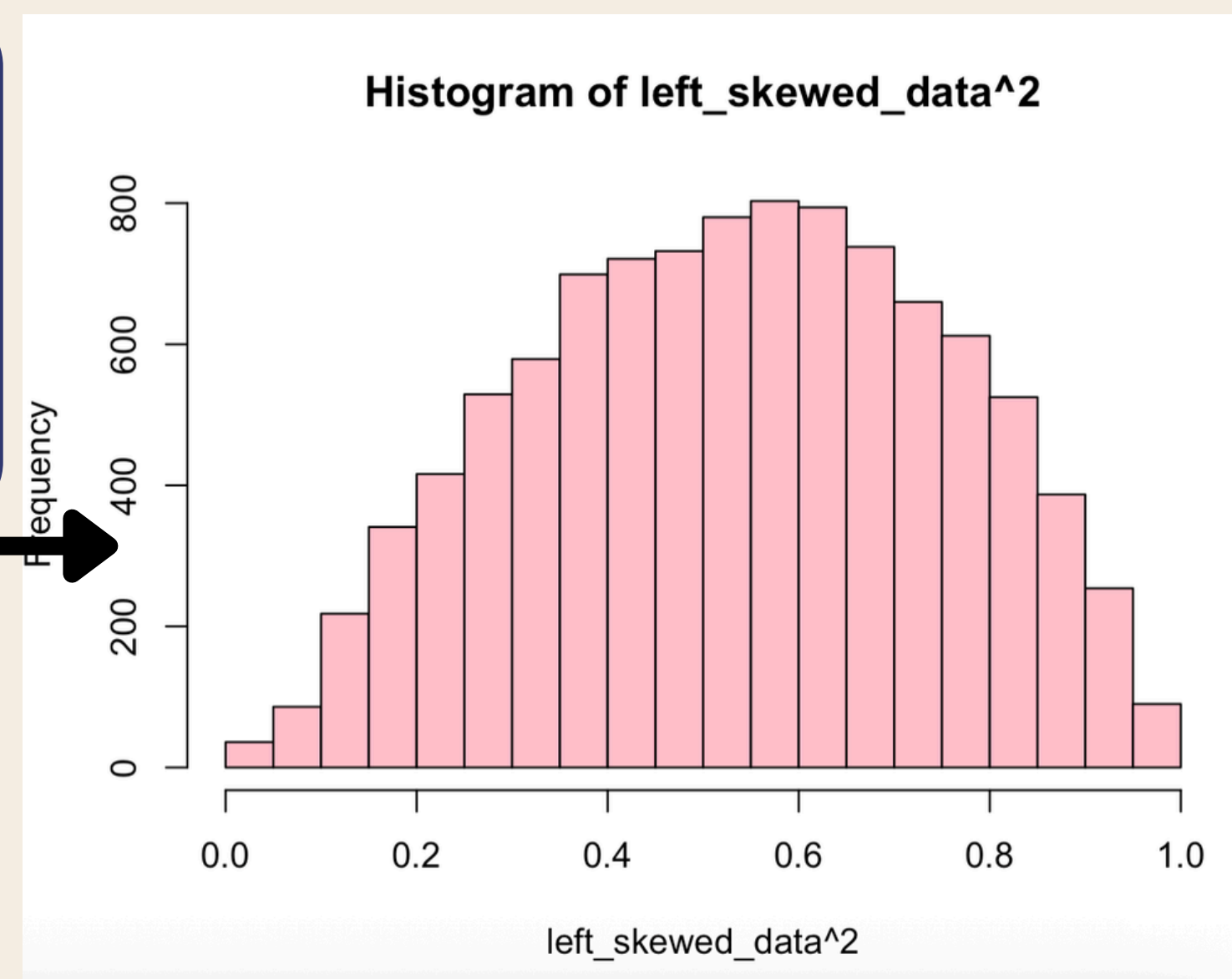


Transforming skewed variables

if left-skewed:

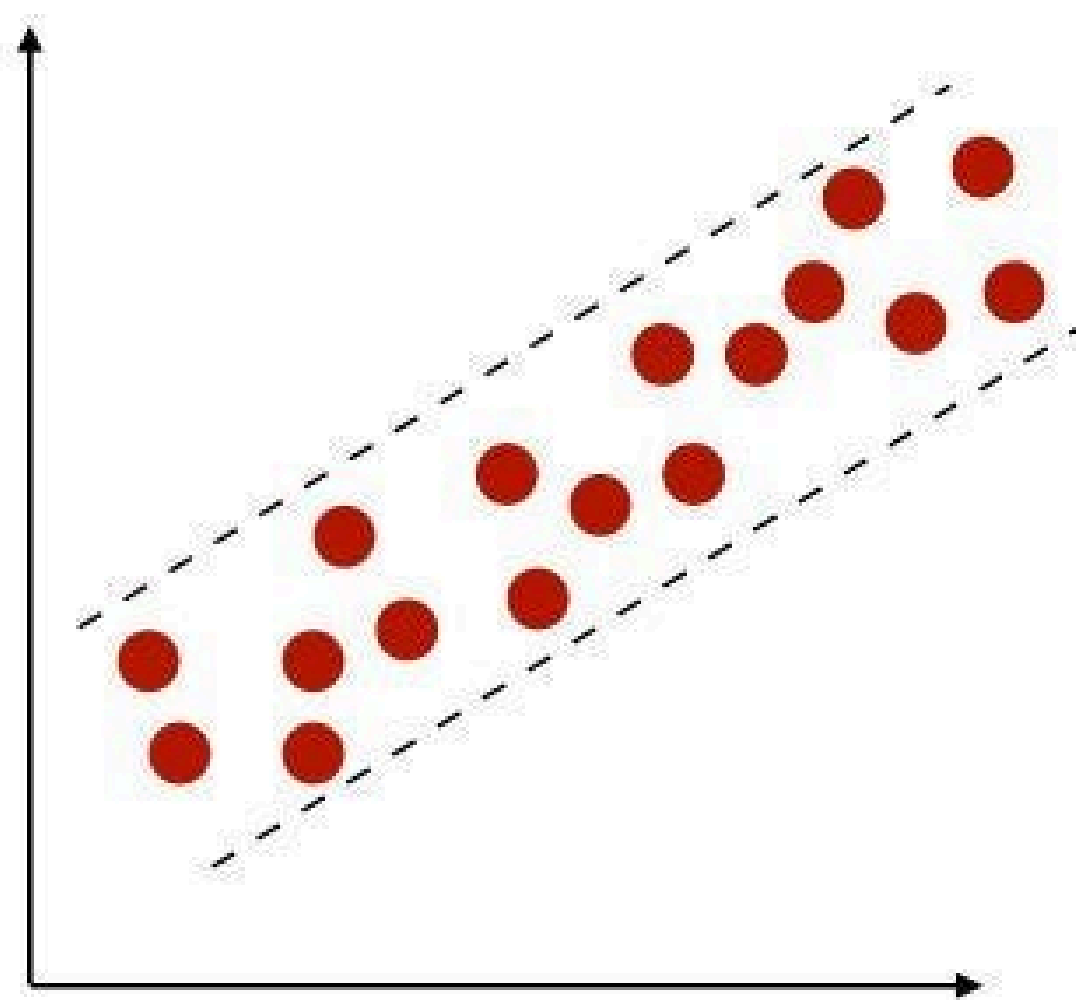


$y \rightarrow y^2$
 $\rightarrow y^3$
 $\rightarrow \exp(y)$

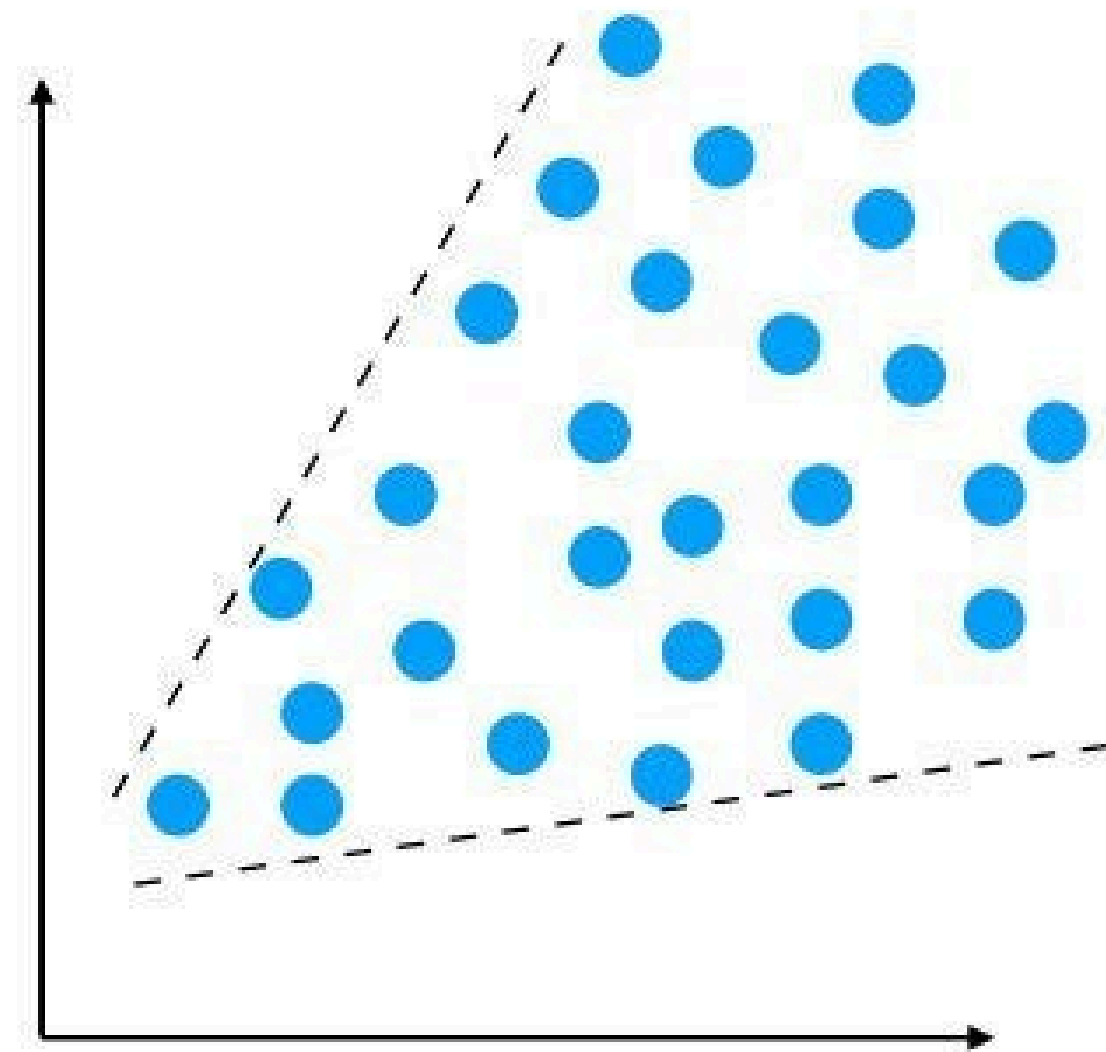


Linear Regression

From last week:



Homoscedasticity



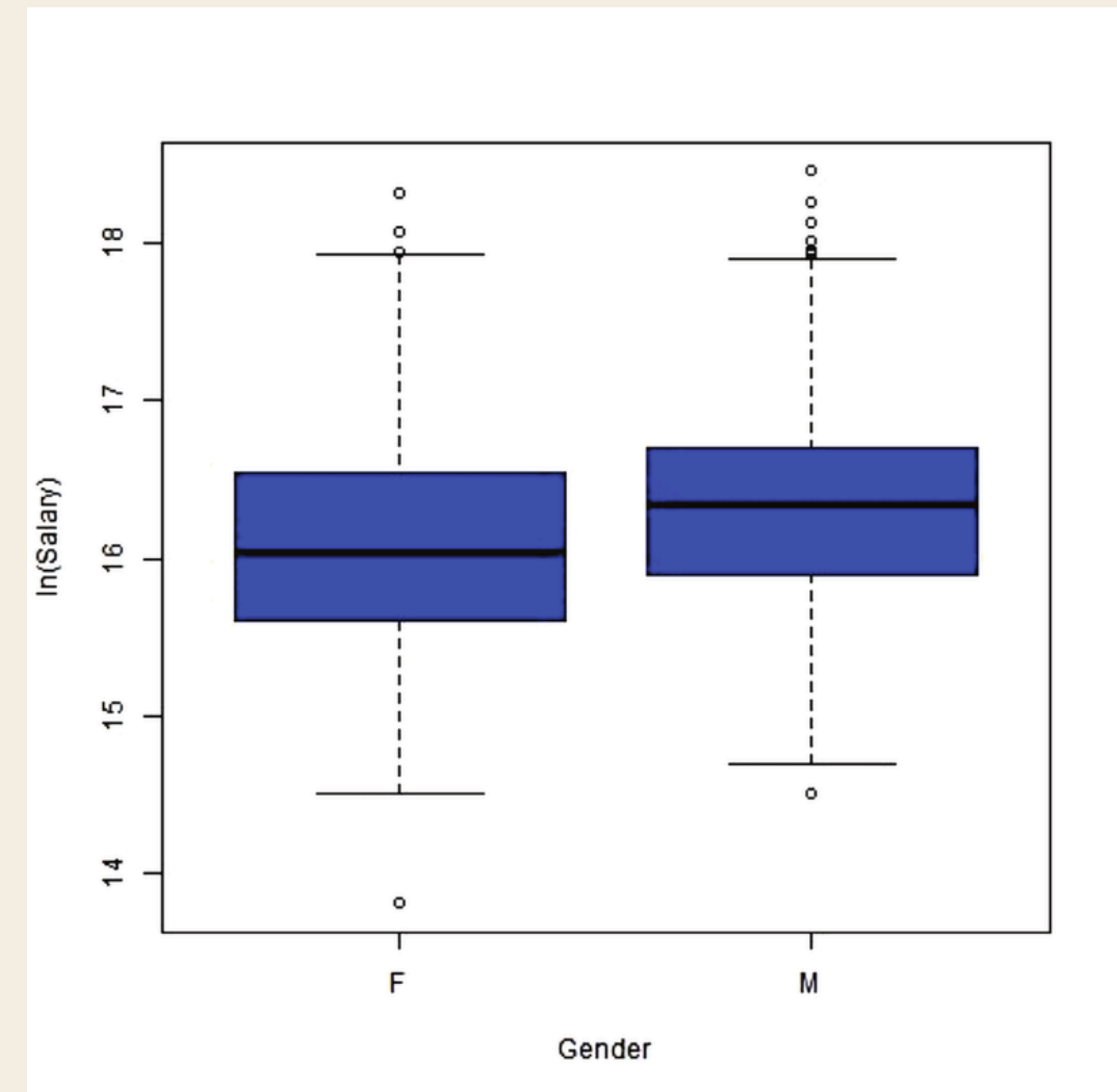
Heteroscedasticity

Linear Regression

What if my x is **categorical**? How can I determine how variability of y is stable when x changes?

- Plot side-by-side boxplot and check if IQR and range is roughly the same across categories

extra: or compare $\sqrt{\text{var}(y \text{ for male})}$ vs $\sqrt{\text{var}(y \text{ for female})}$ ratio less than 2



Evaluating Goodness-Of-Fit

1. R-squared

- The proportion of variation in the response Y that is explained by the fitted model.
- Larger R^2 indicates better model.

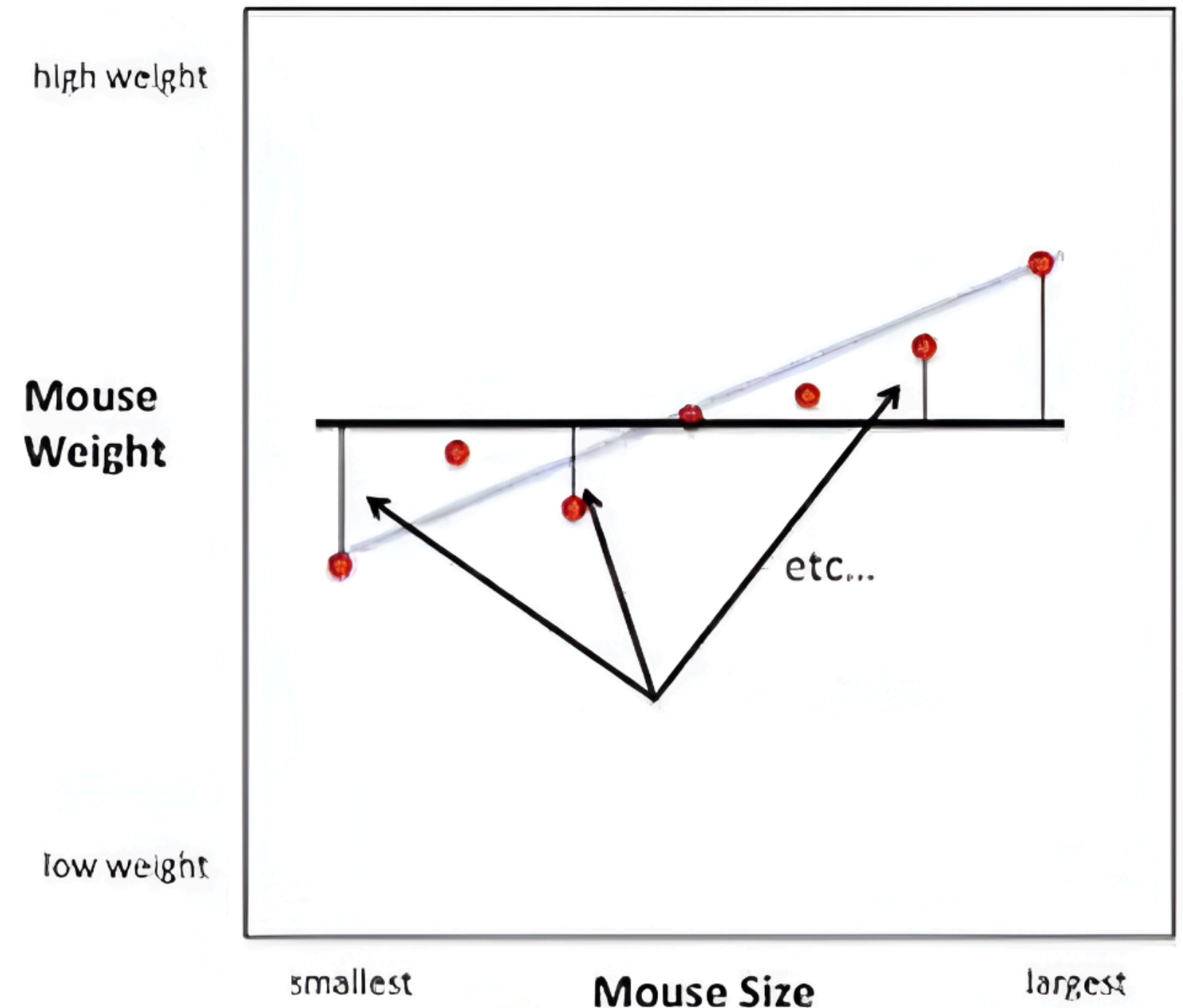
Evaluating Goodness-of-fit

1. R-squared

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

- The proportion of variation in the response Y that is explained by the fitted model.
- Larger R² indicates better model.

Quantifying the difference between the **line** and the mean.
i.e. Calculating R²



Evaluating Goodness-Of-Fit

2. F-test (p-value)

- Does this model (with predictors) explain more variation in y than a model with no predictors?
- p-value: How likely will the model get this F-test value if all predictors were 0 (useless)

$$F = \frac{\text{explained variation}/(k-1)}{\text{unexplained variation}/(n-k)}$$

$$F = \frac{R^2/(k-1)}{(1-R^2)/(n-k)}$$

- High $F \Rightarrow$ the model with predictors reduces error a lot compared to the intercept-only model
- Low p-value (eg. < 0.05) $\Rightarrow$ this improvement is unlikely to be due to random chance

Evaluating Goodness-Of-Fit

Difference between F-statistics and R-squared

R^2 answers:

- How much of the total variation in y is explained by the model?

F-statistic answers:

- Is the explained variation large enough, relative to noise and model size, to conclude that at least one predictor is useful?

Evaluating Goodness-Of-Fit

2. F-test (p-value)

 p-values: What they are and how to interpret them  Copy link

p-values: What they are...



vs 



...and how to interpret them!!!

Watch on  YouTube

<https://www.youtube.com/watch?v=vemZtEM63GY>

lm() function

```
fev_data <- read.csv("Data/fev.csv")
```

```
model <- lm(fev_data$FEV ~ fev_data$height) ❌
```

```
model <- lm(FEV ~ height, data = fev_data) ✅
```

Let's see why together!

summary of model

$$\hat{FEV} = -5.4337 + 5.1971 * height$$

```
> summary(model)
```

Call:

```
lm(formula = FEV ~ height, data = fev_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.73335	-0.26181	-0.00955	0.24283	2.12876

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.4337	0.1820	-29.86	<2e-16 ***
height	5.1971	0.1167	44.54	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4315 on 652 degrees of freedom

Multiple R-squared: 0.7527, Adjusted R-squared: 0.7523

F-statistic: 1984 on 1 and 652 DF, p-value: < 2.2e-16

Interpretation:

If other regressors are fixed, when the height increases by 1 unit, the predicted response increases by 5.1971 units

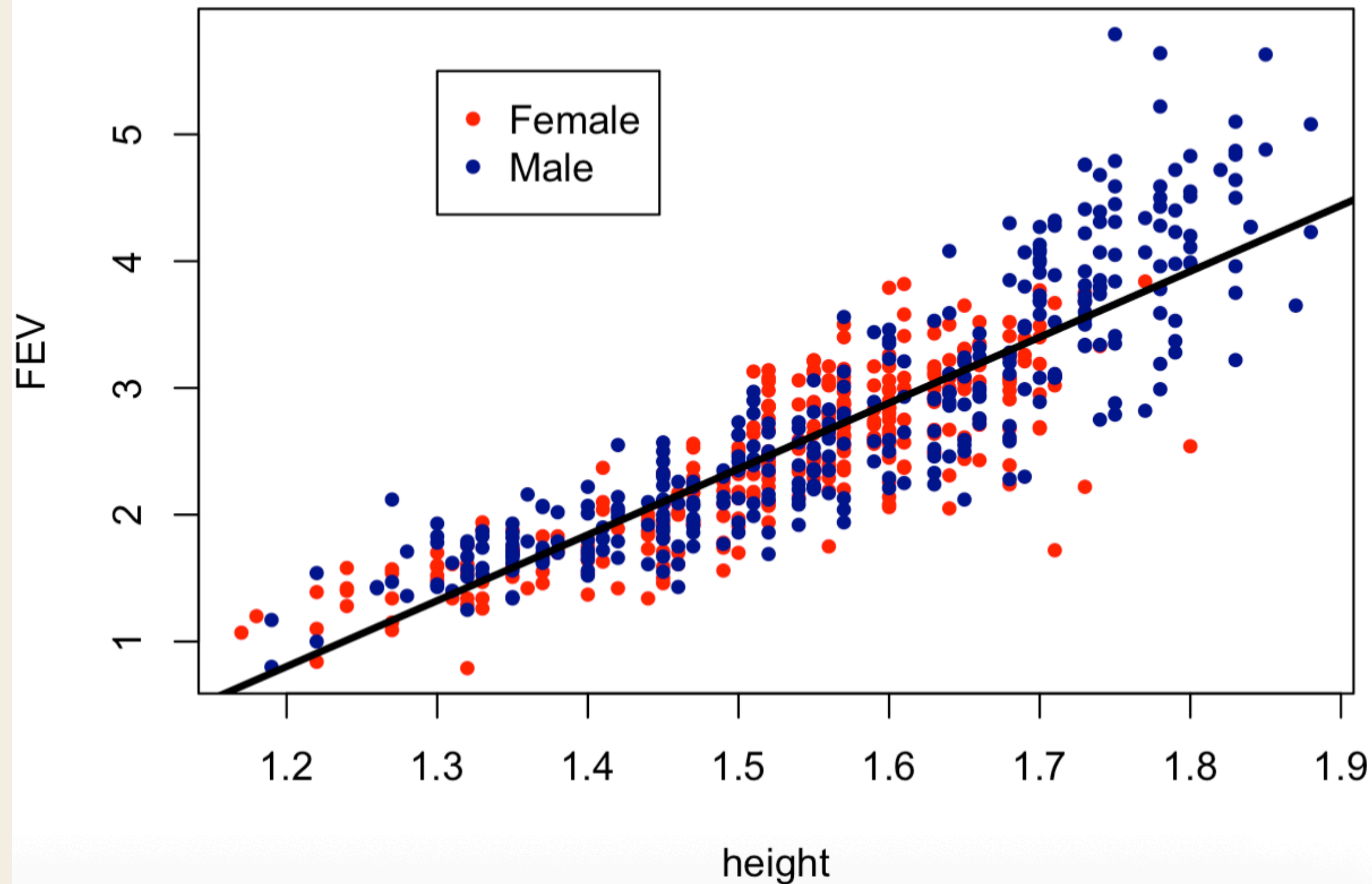
alternatively:

`summary(model)$coef`

Plot Linear Regression Line

Scatterplot of FEV against height

$$\hat{FEV} = -5.4337 + 5.1971 * height$$



predict()

```
model <- lm(FEV ~ height, data = fev_data)
new_data <- data.frame(height = 1.7)
predict(model, newdata = new_data)
```

predict() use the model to predict
output of new data

Output

1
3.401382

predict()

```
model <- lm(FEV ~ height, data = fev_data)
new_data <- data.frame(height = c(1.6, 1.56, 1.7))
predict(model, newdata = new_data)
```

predict() use the model to predict
output of new data

Output

1	2	3
2.881669	2.673784	3.401382

MLR with categorical regressor

```
fev_data$Sex = as.factor(fev_data$Sex)
```

```
model_cat <- lm(FEV ~ height + Sex, data = fev_data)
```

```
summary(model_cat)
```

- `as.factor()` to change it to factor
- `'+'` to add regressor in `lm()`

MLR with categorical regressor

```
> summary(model_cat)
```

Call:

```
lm(formula = FEV ~ height + Sex, data = fev_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.65938	-0.26078	0.00391	0.23506	2.08233

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.39103	0.18068	-29.837	< 2e-16 ***
height	5.12889	0.11711	43.794	< 2e-16 ***
Sex1	0.12313	0.03388	3.634	0.000301 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4275 on 651 degrees of freedom

Multiple R-squared: 0.7576, Adjusted R-squared: 0.7568

F-statistic: 1017 on 2 and 651 DF, p-value: < 2.2e-16

$$\hat{FEV} = -5.39103 + 5.12889 * height + 0.12313 * I(Sex = 1)$$

Interpretation:

When other regressors are fixed, the predicted response of FEV for male (sex = 1) is 0.12313 units larger than that of a female (sex = 0)

ON-SITE QUESTIONS

ON-SITE QUESTION 1,2

A dataset on house selling price was randomly collected ¹, `house_selling_prices_FL.csv`. It's our interest to model how y = selling price (dollar) is dependent on x = the size of the house (square feet). A simple linear regression model (y regress on x) was fitted, called Model 1.

The given data has another variable, NW, which specifies if a house is in the part of the town considered less desirable (NW = 0).

1. Derive the correlation between x and y .
2. Derive a scatter plot of y against x . Give your comments on the association of y and x .

Mostly concepts learnt from last week.
Let's use this as a warm up!

Discuss with your group! Time limit: 4 mins

ON-SITE QUESTION PREP

Step 1: Import and Inspect Data

```
# Import data into R

readLines("Data/house_selling_prices_FL.csv", n=3)

house = read.csv("Data/house_selling_prices_FL.csv")

# Examine data

names(house) # columns
dim(house) # 100 9
head(house)
str(house) # Data types
```

ON-SITE QUESTION 1 PREP

Step 1: Import and Inspect Data

```
> str(house) # Data types
'data.frame':   100 obs. of  9 variables:
 $ House      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Taxes      : int  1360 1050 1010 830 2150 1230 150 1470 1850 320 ...
 $ Bedrooms   : int  3 1 3 3 3 3 2 3 3 3 ...
 $ Baths      : num  2 1 1.5 2 2 2 2 2 2 2 ...
 $ Quadrant   : chr   "NW" "NW" "NW" "SW" ...
 $ NW         : int  1 1 1 0 1 1 1 1 1 1 ...
 $ price      : int  145000 68000 115000 69000 163000 69900 50000 137000 121300 70000 ...
 $ size       : int  1240 370 1130 1120 1710 1010 860 1420 1270 1160 ...
 $ lot        : int  18000 25000 25000 17000 14000 8000 15300 18000 16000 8000 ...
```

The question mentioned this column represents more desirable (NW = 1) or least desirable (NW = 0)

ON-SITE QUESTION 1 PREP

Step 2: Transform NW column to factor

```
house$NW <- as.factor(house$NW)
```

```
> str(house)
'data.frame':  100 obs. of  9 variables:
 $ House      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Taxes      : int  1360 1050 1010 830 2150 1230 150 1470 1850 320 ...
 $ Bedrooms   : int  3 1 3 3 3 3 2 3 3 3 ...
 $ Baths      : num  2 1 1.5 2 2 2 2 2 2 2 ...
 $ Quadrant   : chr   "NW" "NW" "NW" "SW" ...
 $ NW         : Factor w/ 2 levels "0","1": 2 2 2 1 2 2 2 2 2 2 ...
 $ price      : int  145000 68000 115000 69000 163000 69900 50000 137000 121300 70000
 $ size       : int  1240 370 1130 1120 1710 1010 860 1420 1270 1160 ...
 $ lot        : int  18000 25000 25000 17000 14000 8000 15300 18000 16000 8000 ...
```

EXTRA THINKING:

What other columns in `house` can you transform to factor?

ON-SITE QUESTION 1 PREP

Step 3: attach() data

```
> attach(house)
> house$Quadrant <- as.factor(house$Quadrant)
> class(Quadrant)
[1] "character"
```

IMPORTANT:

you must finalize all data type changes before using attach()

attach() simply creates a COPY of the data frame.

so any changes after attach() will not update unless you detach() and attach() again

ON-SITE QUESTION 1 ANSWER

Derive the correlation between x (size of house) and y (selling price).

```
cor(price, size) # 0.76126, quite strong positive correlation
```

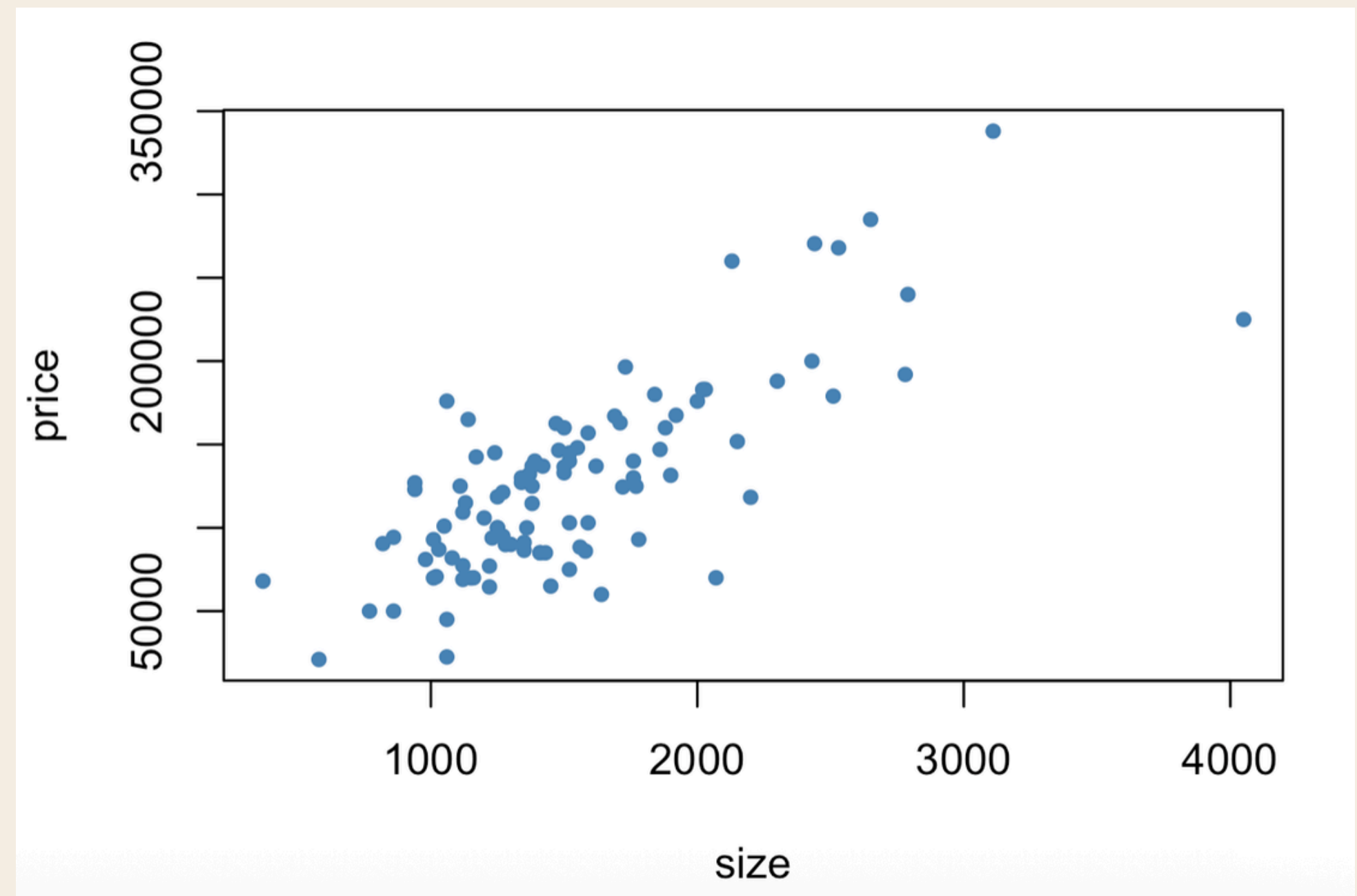
ON-SITE QUESTION 2 ANSWER

Derive a scatter plot of y against x.

```
plot(size, price, pch = 20, col = "steelblue")
```

Comments:

- clear (obvious) association shown.
- association is positive.
- association is quite linear.
- variability of y (the price) is quite stable when x (the size) changes.



ON-SITE QUESTION 3

A dataset on house selling price was randomly collected ¹, `house_selling_prices_FL.csv`. It's our interest to model how y = selling price (dollar) is dependent on x = the size of the house (square feet). A simple linear regression model (y regress on x) was fitted, called Model 1.

The given data has another variable, NW, which specifies if a house is in the part of the town considered less desirable (NW = 0).

3. Derive R^2 of Model 1. Verify that $\sqrt{R^2} = |\text{cor}(y, x)|$. In which situation we can have $\sqrt{R^2} = \text{cor}(y, x)$?

Discuss with your group! Time limit: 4 mins

ON-SITE QUESTION 3 ANSWER

Step 1: Fit Model 1 and get R-squared

```
M1 = lm(price ~ size, data = house)
```

```
summary(M1)$r.squared # 0.57952
```

What other components summary() has other than \$r.squared?

Try: ?summary.lm

ON-SITE QUESTION 3 ANSWER

Step 2: Compare sqrt(R-squared) with absolute correlation

```
root_r_square = sqrt(summary(M1)$r.squared) # 0.7612621  
abs_cor = abs(cor(price, size)) # 0.7612621  
all.equal(root_r_square, abs_cor) # TRUE
```

EXTRA: when comparing two numerical values, use `all.equal(num1, num2)` rather than `num1==num2` to avoid floating-point precision difference

ON-SITE QUESTION 3 ANSWER

Step 2: Compare $\sqrt{R\text{-squared}}$ with absolute correlation

```
root_r_square = sqrt(summary(M1)$r.squared) # 0.7612621
```

```
abs_cor = abs(cor(price, size)) # 0.7612621
```

```
all.equal(root_r_square, abs_cor) # TRUE
```

FAQ: Not required in this course for you to prove it mathematically

Intuitive Idea:

correlation $\rightarrow$ direction + strength of linear relationship

r-squared $\rightarrow$ strength of linear relationship

ON-SITE QUESTION 3 ANSWER

Q: In which situation we can have $\sqrt{R^2} = \text{cor}(y, x)$

When $\text{cor}(y, x) \geq 0$ then in a **simple model** $y \sim x$, we always have $\sqrt{R^2} = \text{cor}(y, x)$.

Note that: $\sqrt{R^2} = |\text{cor}(y, x)|$ only works on simple linear model (1 regressor). Not multiple linear regression.

ON-SITE QUESTION 4,5

4. Form a model (called Model 2) which has two regressors (x and NW). Report the coefficient of variable NW in Model 2. Interpret it.
5. Estimate and report the price of a house where its size is 4000 square feet and is located at the more desirable part of the town.



Discuss with your group! Time limit: 7 mins

ON-SITE QUESTION 4 ANSWER

Q: Form a model (called Model 2) which has two regressors (x and NW).

```
M2 = lm(price ~ size + NW, data = house)
```

ON-SITE QUESTION 4 ANSWER

Q: Report the coefficient of variable NW in Model 2. Interpret it.

```
> summary(M2)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-15257.51385	11908.297056	-1.281251	2.031602e-01
size	77.98513	6.209194	12.559622	4.690632e-22
NW1	30569.08729	7948.742292	3.845777	2.149752e-04

For two houses of the same size (fixed x), the price of the house in the more desirable part $*NW = 1*$ is predicted to be **\$30569.1** more than the one in the less desirable part $*NW = 0*$ (the reference).

ON-SITE QUESTION 4 ANSWER

Extra: Can I change my reference to be NW=1 instead?

Yes! Using `relevel()`

```
house$NW <- relevel(factor(house$NW), ref = "1")
```

```
M3 <- lm(price ~ size + NW, data = house)
```

```
summary(M3)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	15311.57343	10200.456838	1.501067	1.365863e-01
size	77.98513	6.209194	12.559622	4.690632e-22
NW0	-30569.08729	7948.742292	-3.845777	2.149752e-04

ON-SITE QUESTION 5 ANSWER

Q: Estimate and report the price of a house where its size is 4000 square feet and is located at the more desirable part of the town.

size = 4000, NW = 1

```
new = data.frame(size=4000, NW = "1")
```

```
predict(M2, newdata = new) # 327252.1
```

```
# The mean price of a house with size x = 4000 and NW = 1 is $327252.1
```

OFF-SITE QUESTIONS

We're going to do

Preference on to teach mainly using:

👍 45

Code



Slides



Code!

OFF-SITE QUESTION 1 ANSWER

Q: Write your own function in R, called simple, to derive the intercept β_0 and the slope β_1 of Model M1

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}}$$

Formula of the estimated coefficients given in slide 31 of Topic 3.

OFF-SITE QUESTION 1 ANSWER

The diagram shows the formula for the slope coefficient $\hat{\beta}_1$ in linear regression. The formula is:
$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right) \left(\sum_{i=1}^n x_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}}$$
 Annotations with arrows point from text labels to specific parts of the formula:

- $\text{sum}(y^*x)$ points to $\sum_{i=1}^n y_i x_i$
- $\text{sum}(y)$ points to $\sum_{i=1}^n y_i$
- $\text{sum}(x)$ points to $\sum_{i=1}^n x_i$
- $\text{sum}(x^2)$ points to $\sum_{i=1}^n x_i^2$
- $\text{sum}(x)^2$ points to $\left(\sum_{i=1}^n x_i\right)^2$

$$\text{beta1} = (\text{sum}(y^*x) - (\text{sum}(y)*\text{sum}(x)/\text{length}(x))) /$$
$$(\text{sum}(x^2) - \text{sum}(x)^2 / \text{length}(x))$$

OFF-SITE QUESTION 1 ANSWER

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right) \left(\sum_{i=1}^n x_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}}$$

Diagram illustrating the components of the formula for $\hat{\beta}_1$:

- $\sum_{i=1}^n y_i x_i$ is labeled $\text{sum}(y*x)$.
- $\sum_{i=1}^n y_i$ is labeled $\text{sum}(y)$.
- $\sum_{i=1}^n x_i$ is labeled $\text{sum}(x)$.
- $\sum_{i=1}^n x_i^2$ is labeled $\text{sum}(x^2)$.
- $\left(\sum_{i=1}^n x_i\right)^2$ is labeled $\text{sum}(x)^2$.

$$\text{beta1} = (\text{sum}(y*x) - (\text{sum}(y)*\text{mean}(x))) /$$
$$(\text{sum}(x^2) - \text{sum}(x)*\text{mean}(x))$$

**EXTRA
KNOWLEDGE**

POLYNOMIAL LINEAR REGRESSION

```
polynomial_model <- lm(FEV ~ height + I(height^2), data = fev_data)
summary(polynomial_model)
```

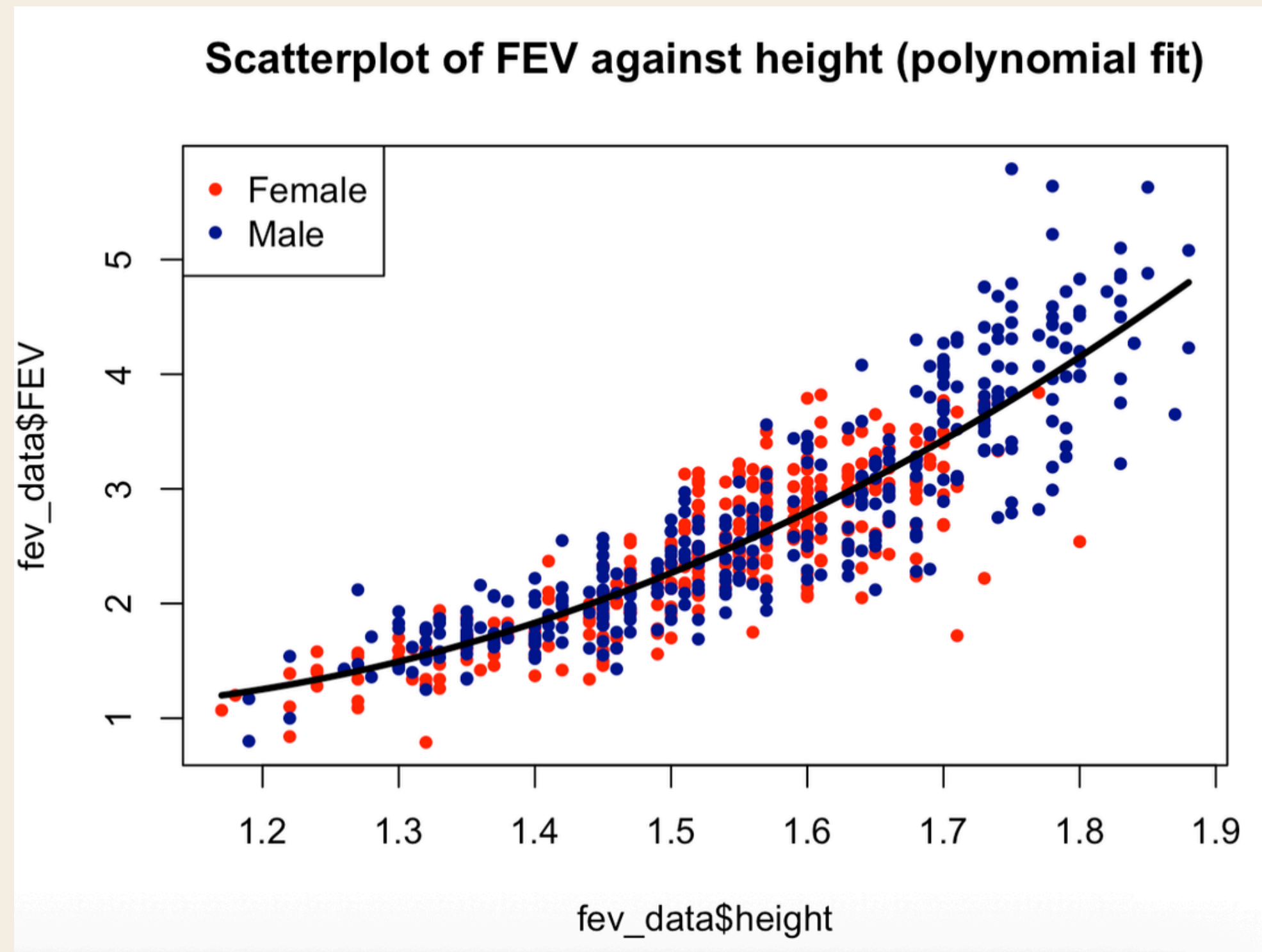
```
> summary(polynomial_model)$coef
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.912404	1.507875	3.921018	9.752612e-05
height	-9.691843	1.968586	-4.923251	1.080072e-06
I(height^2)	4.840851	0.639011	7.575537	1.232110e-13

$$\text{FEV_hat} = 5.912 - 9.692(\text{height}) + 4.841(\text{height}^2)$$

POLYNOMIAL LINEAR REGRESSION

$$\text{FEV_hat} = 5.912 - 9.692(\text{height}) + 4.841(\text{height}^2)$$



POLYNOMIAL LINEAR REGRESSION

Won't come out for exams! Just
may be useful for your projects.

THANK YOU!



Weekly Slides & Code will be shared on Github (★ star it for bookmark):

<https://github.com/kr7001/DSA1101-teaching-materials>